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Plastics, Rubber and Composites
Macromolecular Engineering

ISSN: 1465-8011 (Print) 1743-2898 (Online) Journal homepage: http://www.tandfonline.com/loi/yprc20

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PLASTICS, RUBBER AND COMPOSITES, 2016
http://dx.doi.org/10.1080/14658011.2016.1245807

Durability and degradation mechanism of GFRP bars embedded in concrete beams with cracks

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ABSTRACT

This paper reports an experimental study on the durability performance of glass fibre-reinforced polymer (GFRP) bars embedded in concrete beams with cracks. Accelerated tests were performed by immersing the beams in an alkaline solution at 60°C and in tap water at 23°C. Tensile tests were conducted to evaluate the residual tensile properties of the aged GFRP bars. Moreover, scanning electron microscopy, Fourier transform infra-red spectroscopy and differential scanning calorimetry were employed to observe and analyse potential deterioration of the fibres, matrix and fibre/matrix interface. The results showed that the tensile strength of the GFRP bars embedded in concrete beams decreased in all the different environments tested, and most of the tensile strength losses occurred during the initial stage. Actually, compared to the accelerated aging environment, the degradation of the tensile strength of GFRP bars embedded in concrete in the real-world environment is minor because the concrete, which has superior barrier properties, acts like a protective cover for the GFRP bar. Based on the test results, it can be concluded that a crack in concrete may increase the environment effect on the durability performance of the GFRP bar, but cannot change the mechanisms of mechanical degradation of the GFRP bar. In addition, the degradation rate of the GFRP bar was accelerated by sustained flexural loading. Consequently, it is important to restrain and alleviate the combined interaction of a harsh environment and sustained loading in practical engineering to enhance the durability of GFRP bars.

ARTICLE HISTORY
Received 10 August 2016
Accepted 2 October 2016

KEYWORDS
Durability; GFRP; Degradation mechanism; Sustained load; Microstructure

Introduction

Fibre-reinforced polymer (FRP) bars, which are used as an alternative to steel reinforcement in concrete structures to resolve corrosion issues, have become increasingly popular, particularly because of the high cost of maintenance generally associated with critical infrastructure [1–3]. The unique mechanical and physical properties of FRP composites, such as high strength, low weight, non-corrosive nature and ultraviolet resistance, make FRP bars a viable alternative to steel reinforcement [4,5]. However, the lack of long-term performance data for real-world concrete environments is cited as a major obstacle hindering wider acceptance of FRP materials in infrastructure applications. Numerous studies [6–8] have been performed on the durability of GFRP bars under different harsh environments. Chen et al [4] studied the durability of bare GFRP bars and bars embedded in concrete immersed in different solutions at different temperatures and times of exposure. The authors found that bare GFRP bars and the bars embedded in concrete showed a significant strength loss when exposed to simulated environments, especially for solutions at the high temperature of 60°C. Dejke [9] discussed the concrete environment and its effect on GFRP in terms of internal and external aggressive conditions that may affect the GFRP durability. Nevertheless, these studies were all focused on evaluating the durability of GFRP bars under severe environments without sustained loading, which is different from the service conditions.

Several studies (Almusullam and Al-Salloum; [10] Nkurunziza et al [11]) have addressed the durability of GFRP bars embedded in concrete beams subjected to axial loading. While concrete beams are mostly subject to flexural load in practical engineering, the degree of chloride penetration in concrete that experience flexural bending load is quite different from a beam that experiences axial loading. Furthermore, the influence of cracks in the concrete is ignored, which is different from the concrete environments. In fact, a concrete structure usually has micro-cracks in service because of the characteristics of concrete materials and the effect of other adverse factors such as poor construction and overloading. The micro-cracks in concrete create stress concentrations when the beams are subjected to flexural loading, which accelerates the emergence and propagation of micro-cracks [12,13]. These micro-cracks provide additional channels for the external environment media to attack the fibres, which adversely affect the durability of the GFRP bars. However, very limited studies [14,15] have been

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conducted on the durability of GFRP bars embedded in concrete beams with cracks.

In the present study, accelerated aging tests of GFRP bars embedded in concrete beams under service loads with cracks were performed. After reaching the specified corrosive period, the bars were extracted from the beams. Tensile testing was conducted to evaluate the residual tensile properties of the aged GFRP bars. Scanning electron microscopy (SEM), Fourier transform infra-red (FTIR) spectroscopy and differential scanning calorimetry (DSC) were employed to characterise the deterioration of the fibre, matrix and fibre/matrix interface to further understand the degradation mechanisms of the GFRP bars embedded in concrete beams.

Experimental programme

Raw materials and specimens

The GFRP bar used in the study was composed of 72% glass fibre and 28% vinyl ester resin, and manufactured using a pultrusion process by Nanjing Feng Hui Composite Materials Co., ltd, China (as shown in Figure 1). The surface of the GFRP bar was ribbed with a nylon laminate during the pultrusion. The mechanical and physical properties of 10-mm diameter GFRP bars, as provided by the manufacturer, are summarised in Table 1. The bar's circular cross-section is manufactured by a process that couples pultrusion with sand coating along the external surface of the bar.

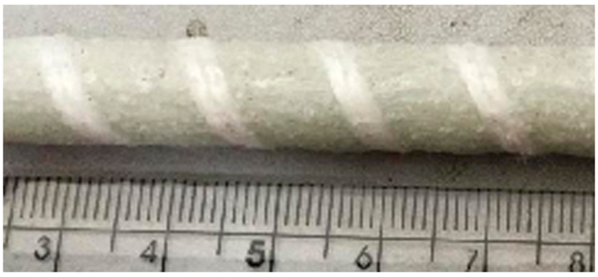

Figure 1. GFRP bar used in tests.

Table 1. Mechanical properties of GFRP bar (as provided by the manufacturers).

Property	Unit	Value
Density	g cm^{-3}	2.20
Tensile strength	MPa	1130
Elastic modulus	GPa	51.6
Elongation at break	%	1.35
Poisson's ratio	…	0.25

The reinforced concrete specimens were rectangular plain concrete beams with a length of 1100 mm, width of 80 mm and depth of 110 mm (as shown in Figure. 2), with a single longitudinal GFRP reinforcement bar centred at 30 mm from the bottom of the section, and no shear reinforcement was used in the beams. Concrete beams were cast according to code for design of concrete structures (GB50010-2010), with 28 days target strength of 30 MPa. To minimise the influence of the end-effects during conditioning of the beams, a commercial concrete sealant was applied to the ends of the concrete beam and a high-grade epoxy was applied to the cut ends of the bars exposed at the ends of the beams.

Test setup and procedure

A spring-bracket assembly was fabricated by the research team to apply the sustained flexural loading and to produce a stable crack, the assembly consisted of steel frames, long threaded bolts with nuts and steel springs with a spring coefficient is 40 N mm^{-1}. The beams were placed back to back, the inner two load points were located 200 mm from the outer load points, while the two outer load points were located 100 mm inward from the ends of the beam, the sketch illustrating of the sustained loading setup is shown in Figure 3. Preliminary four-point flexure tests were performed to determine the ultimate bending moment of the beams and provide the original design data for pre-cracked load. Flexural cracks were introduced in the beams by subjecting them to the pre-cracked load. Failure of four-point flexure tests occurred in the shear/compression mode at one of the inner load points at a mean applied moment of 0.8 kN m. Beams were pre-cracked at a bending moment of 0.39 kN m. A pair of beams was not pre-cracked as a reference, and the rest of beams were pre-cracked by four-point flexure tests after a specified time.

To simulate the service condition under sustained loading, four-point flexural loads corresponding to 20 and 40% of the measured ultimate strength of the GFRP bars were applied and sustained by the spring-bracket assembly. The sustained load should prevent creep rupture of the concrete continuous beam [16]. The steel spiral springs were periodically monitored to ensure the cross-section flexural moment within the given error range.

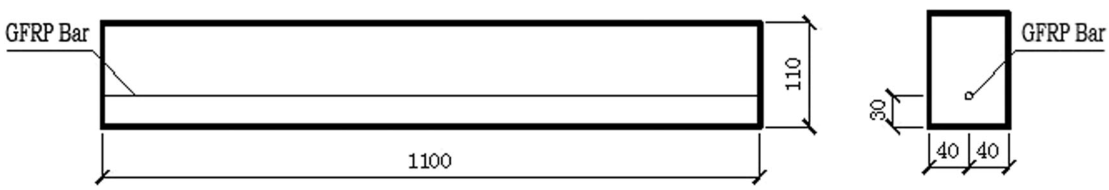

Figure 2. Geometry of specimen.

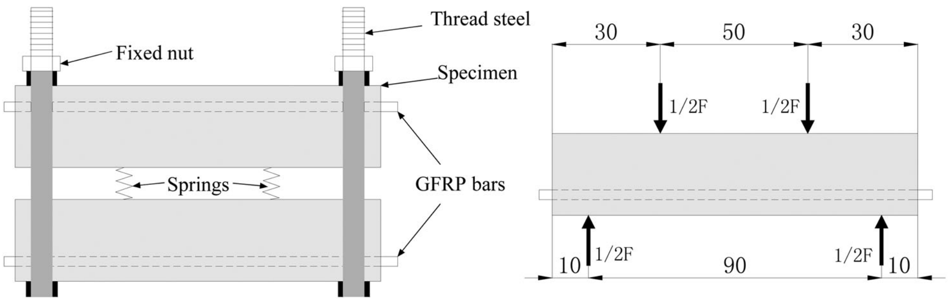

Figure 3. Sketch illustrating typical GFRP bars embedded in concrete beams.

Environmental conditions

To evaluate the durability of GFRP bars embedded in concrete beams with cracks in different environments, three types of the ambient environmental conditions were used: (1) indoor environment (23°C, 70% relative humidity); (2) tap water (23 ± 2°C) environment; and (3) alkaline solution (60 ± 2°C) with a pH value of 13.5. The exposure time for specimens was 6, 12 and 18 months, respectively.

In uncracked regions, the concrete acted as a protective role for the GFRP bar. The crack will provide a channel for the corrosive medium to enter the inside of the beams and to generate different degrees of damage to the components in the crack regions. Therefore, it is necessary to set up two groups of beams, one set with pre-cracks and the other without pre-cracks, to highlight the influence of the crack on the durability performance of the GFRP bar. Four beams were conditioned in each type of environmental condition. A summary of the test programme is presented in Table 2.

Tensile test

After the end of each accelerated corrosion period, the beams were unloaded and the GFRP embedded bars in the tested beams were carefully extracted for tensile tests. The overall length of each bar was 1100 mm. To prevent shear failure on both ends of the GFRP bar, a 250-mm long steel pipe anchorage was fixed on the GFRP bar ends. The tensile tests were conducted in a SHT4106-G type microcomputer controlled electro-hydraulic servo universal testing machine according to the relevant requirements of ACI440.3R-04 B2. The applied load for each bar was recorded during the test with a data acquisition system monitored by a computer.

Microstructure observations

SEM observations and image analysis were performed to observe the microstructure of the specimens in tap water and alkaline solution for 18 months. All specimens observed in the SEM were cut, polished and coated with a thin layer of gold-palladium, using a vapour-deposit process. After coating the surfaces, microstructural observations on the longitudinal surfaces were performed using a JSM-5610LV SEM. These observations were conducted to see the potential degradation of the glass fibres or interfaces, if any.

Fourier transform infra-red spectroscopy

FTIR spectra were recorded using a Nicolet 6700 spectrometer equipped with an attenuated total reflectance device. Fifty scans were routinely acquired with an optical retardation of 0.25 cm to yield a resolution of 4 cm^{-1}. This analysis was performed to determine if hydrolysis reactions occurred in the polymer resin, which can lead to a substantial loss of mechanical properties.

Differential scanning calorimetry

The glass transition temperature (T_g) is an important parameter for analysing the durability behaviour of GFRP bars. A PYRIS1 DSC was used to measure the T_g of the GFRP bars unconditioned bars and aged in

Table 2. Test programme.

Environment	Process method	Sustained load level	Name	Aging (month) 6	12	18
Unconditioned	…		RE	4	4	4
Tap water	Without pre-crack	20%	TW20	4	4	4
	With pre-crack	20%	TPW20	4	4	4
	With pre-crack	40%	TPW40	4	4	4
Alkaline solution	Without pre-crack	20%	AS20	4	4	4
	With pre-crack	20%	APS20	4	4	4
	With pre-crack	40%	APS40	4	4	4

Note: The number '4' represents the specimen number of each group.

a 60°C alkaline solution for 18 months. Specimens weighing 10–13 mg were cut from different GFRP samples. The measurements were made at a constant heat rate of 10°C min^{-1}, from 30 to 200°C in a nitrogen atmosphere with a purge rate of 20 mL min^{-1}. Each sample was scanned twice and the T_g was determined from the second scan.

Results and discussion

Degradation of tensile properties of the GFRP bars

The tensile test of the unconditioned and preloaded GFRP bars showed an approximately linear behaviour up to failure. The specimen failed because the fibres ruptured. The failure was accompanied by the delamination of fibres and resin. A similar, but less catastrophic failure was observed for the GFRP bars embedded in the concrete beams with cracks. No chemical deposit was observed on bar surface before testing.

The variation of average tensile strength retention and losses of the GFRP bars in different environmental conditions is shown in Figures 4 and 5. It can be seen from Figure 5 that, at the end of 18 months of conditioning, the maximum tensile strength losses of 11.4, 12.7,15.8, 20.3, 22.1 and 24.6% were observed in the TW20, TPW20, TPW40, AS20, APS20 and APS40 environments, respectively. However, for the unconditioned specimens in the indoor environment, a slight reduction in the tensile strength was recorded after 6, 12 and 18 months, and the residual strengths were approximately 98.0, 97.4 and 96.2% (losses of 2.0, 2.6 and 3.8%), respectively. This result shows that the GFRP bars have better durability performance in dry concrete than that in moist concrete. This observation is consistent with studies in the literature, [17,18] which conclude that moisture is the main

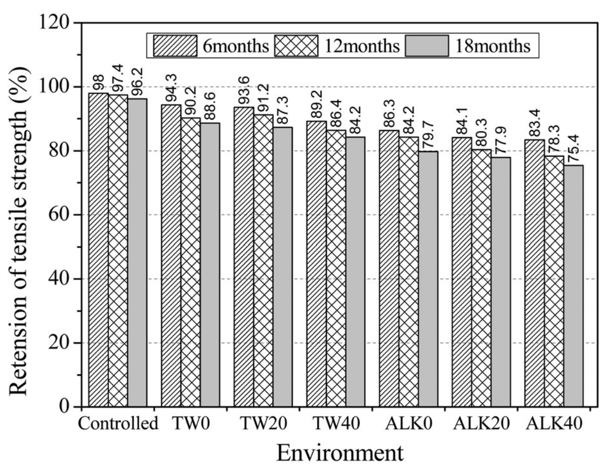

Figure 4. Retention of tensile strength after 6, 12 and 18 months in different environments.

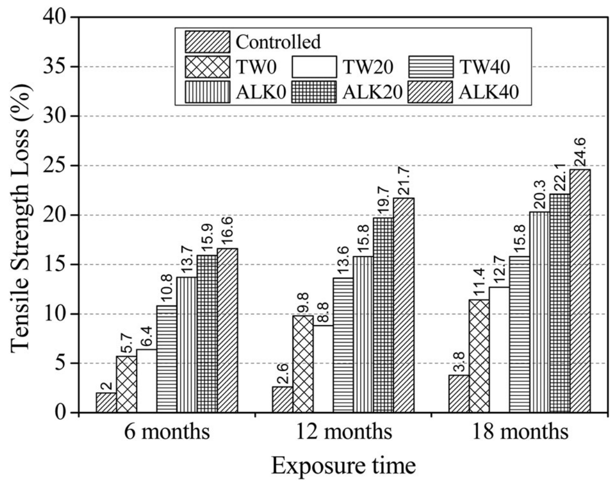

Figure 5. Tensile strength loss of GFRP bars after 6, 12 and 18 months in different environments.

parameters that affect the durability of the composite materials. It is well known that moisture absorbed by the resin matrix induces stresses in the materials that consequently damages the fibres, matrix and their interface, and decreases the strength of the GFRP bars with time [11]. Moreover, moisture can also act as a plasticiser disrupting the van der Waals bonds in the polymer chains, which produces fibre–matrix debonding [18,19].

For the TPW40 specimen, the strength loss was 10.8% after 6 months of exposure. As the exposure time increased, the strength loss slightly increased to 13.6 and 15.8% after 12 and 18 months, respectively, which indicates that most of losses occurred in the first 6 months. Compared to the TW20 and TPW20 specimens, increasing the load level to 40% resulted in faster degradation rates of the tensile strength of the bars. A possible reason for this behaviour is the presence of micro-cracks in the matrix under the presence of stress, which result in further absorption of corrosive ions into the GFRP bar. The recorded strength losses of the APS40 specimen were similar to those obtained in the TPW40 specimen. After 6, 12 and 18 months of exposure, the strength loss was approximately 16.6, 21.7 and 24.6%, respectively. Note that most of the tensile strength losses were recorded during the initial stage and almost no significant losses were recorded after 12 and 18 months. This could be attributed to the formation of a thin layer of reactant on the bars.

Figure 6 shows the average residual tensile modulus of the tested GFRP bars after 6, 12 and 18 months in different environment. It is observed that the environmental conditions and sustained flexural loading did not have significant influences on the tensile modulus of the GFRP bars. For all the environments, a slight decrease ranging between 0 and 5% was observed. These results are in good agreement with the results presented in other studies [12,18].

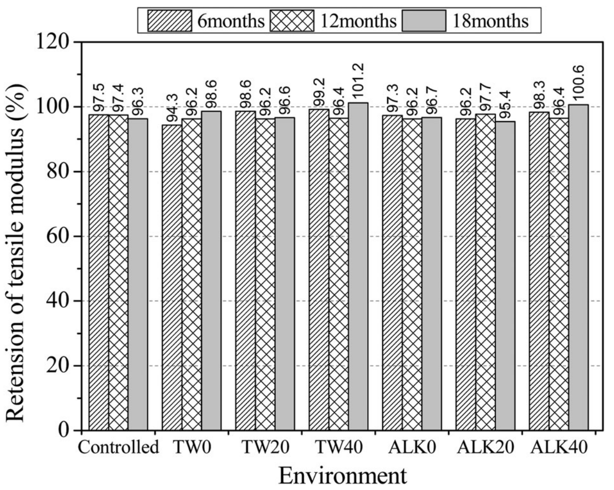

Figure 6. Retention of tensile modulus of GFRP bars after 6, 12 and 18 months in different environments.

Degradation of glass fibre of the GFRP bar

Micrographs of the GFRP bars before and after aging in concrete beams with and without pre-cracks are shown in Figure 7. The GFRP bars embedded in the concrete beams with pre-cracks had more residual matrix on the fibres after failure and showed smooth fibres after debonding and failure, which indicated weaker adhesion due to the presence of weak ester groups. The profile of fibre appeared larger defects and the surface formed the obvious pitting and etching and leaching, and fracture even occurred in part of the fibre when the beams were in alkaline solution for 18 months. At the same time, there are more micro-cracks in the matrix and obvious peeling off between fibres and resin.

Equation (1) shows the reaction that causes the glass fibre dissolution in water. In this process, alkali ions are extracted out of the glass fibre, and the leaching process continues as long as the alkalis are available in the glass fibre [20]. Hydroxides produced by the leaching process increase the pH, and once the pH exceeds nine, it affects the Si–O–Si network, according to equation (2). In alkaline environments, the glass fibre can be degraded by silica dissolution with hydroxides, as shown by equation (2). The first reaction products of equation (2) can accelerate the reaction with water molecules to produce alkali ions, which accelerates equations (1) and (2) [8]. Alkali ions destroy the backbone of the glass molecules by breaking the Si–O–Si network into dissolvable products, which leads to the formation of longitudinal cracks and peeling off, while dissolution and leaching of dissolvable products result in pitting and etching [21]. The reaction between the glass fibre and alkali ions will lead to degradation on the mechanical properties of the glass fibre, and hence a reduction of the tensile properties of the GFRP bars embedded in concrete beams after aging.

$$\mathrm{Si} - \mathrm{O}^- + \mathrm{H_2O} \rightarrow \mathrm{Si} - \mathrm{OH} + \mathrm{OH}^- \qquad (1)$$

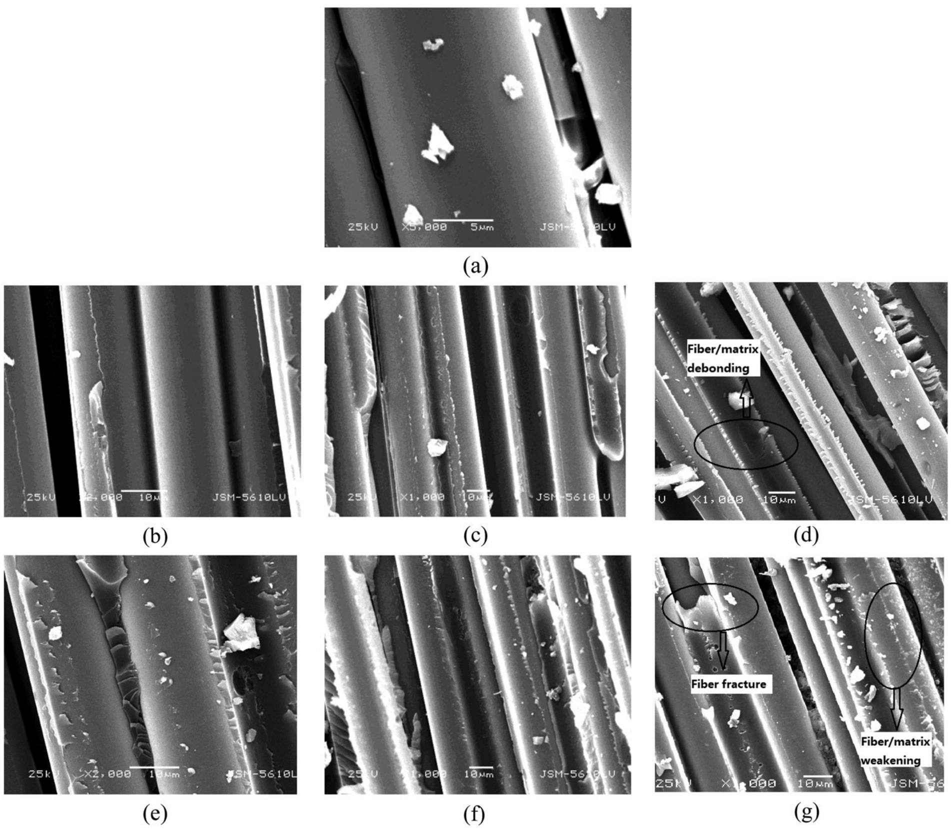

Figure 7. Typical SEM images for the longitudinal of GFRP bars: (a) RE (×5000); (b) TW20-18 months (×2000); (c) TPW20-18 months; (×1000) (d) TPW40-18 months (×1000); (e) AS20 -18 months (×2000); (f) APS20-18 months (×1000); (g) APS 40-18 months (×1000).

$$Si-O-Si\ +OH^- \rightarrow Si-O-(\text{solution})+ Si-OH\ (\text{solid}) \quad (2)$$

Figure 7 shows the SEM micrographs of the examined GFRP samples at high magnifications. As shown in Figures 7(b), (c) and (d), there was almost no deterioration in the glass fibre, while the polymer matrix around the glass fibres in both the tap water and alkaline solution specimens were seriously affected by the environment. It was also seen that no significant evidence of degradation was observed for a sustained flexural loading level less than 20% of the ultimate bending moment, and this result was consistent with the tensile test analysis. When the sustained flexural load level increases to 40%, the glass fibre of the GFRP bars embedded in concrete beams with cracks exposed to an alkaline solution appeared to split. However, an obvious difference between the micrographs of the beams with and without pre-cracks is not found.

Degradation of fibre/matrix interface of the GFRP bar

The micrographs in Figure 7 show the fibre/matrix interface for *a* a reference sample, (b)–(d) GFRP bar embedded in a concrete beam aged in tap water for 18 months at 23°C and (e)–(g) GFRP bar embedded in a concrete beam aged in an alkaline solution for 18 months at 60°C. Exposure of the specimen to tap water (TW20, TPW20 and TPW40) did not affect the fibre/matrix interface, which is in agreement with Robert et al [18] reported. While in the specimen exposed to the alkaline solution (AS20, APS20 and APS40), the bonds between the fibre and matrix becomes loose, and as the sustained flexural load level increased, the interfacial debonding become more significant, which indicated a weakening of the interfacial bonding after aging. This is a major reason for the reductions in the tensile properties of the GFRP bar embedded in concrete beams.

Degradation of polymer matrix of the GFRP bar

When a hydrolysis reaction occurs, new hydroxyl groups are formed and the corresponding infra-red bands increase. Different vibration frequency close to groups of the same type of molecules, and the strong absorption band of the hydroxyl groups is located between 3300 and 3600 cm^{-1}. Figure 8 shows the FTIR analysis of the reference and APS40 specimen after exposure for 18 months. Changes in the peak intensity are quantified by determining the ratio of the OH^- peak to the resin's carbon hydrogen stretching peak, which is not affected by the conditioning [22]. However, the hydroxyl peak did not show any significant changes and the experimental spectra did not show any new characteristic absorption peak. This

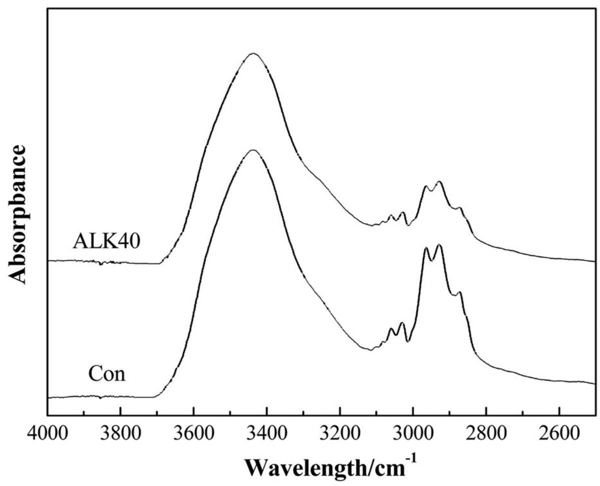

Figure 8. FTIR spectra for RE and APS40 samples.

observation led to the conclusion that the resin matrix of the GFRP bar embedded in concrete beams with cracks did not form a new group after resin ester bond rupture in an alkaline solution at 60°C.

DSC was used to measure the T_g of the GFRP bars embedded in concrete beams with and without pre-cracks. Table 3 presents the T_g values for the first and second heating scans of the GFRP bar samples. The first scan determines the difference in the T_g between the reference and conditioned specimens. A decrease in T_g for the conditioned specimens is indicative of a plasticising effect or chemical degradation. A low T_g of the conditioned specimens after the second scan is indicative of a nonreversible chemical degradation, such as hydrolysis [22]. It can be seen from Table 3 that the T_g values of the second scan for all tested samples are higher than that of the first scan. This shift indicates that the samples had not been completely cured and a post-curing phenomenon occurred during the second heating scan. The T_g values of the AS20 and APS40 specimens compared with the reference specimen decreased by 2.4 and 4.1%, respectively, in the first scan. This result could be explained by the fact that aging irreversible chemical reaction (such as hydrolysis) of the matrix resin has occurred, however, this has little effect on the thermal performance of the AS and APS specimens.

The T_g of the GFRP bar did not change after exposures shows little or no long-term effect of water or hydroxyl ingress on the bulk resin matrix material. One can conclude that no major degradation occurred

Table 3. The glass transition temperature of GFRP bar under different environments.

Name	Temperature (°C)	Duration (m)	T_g run1 (°C)	T_g run2 (°C)	Cureratio (%)
RE	…	…	123	125	98.4
AS20	60	18	120	124	96.8
APS40	60	18	118	124	95.2

at the polymer level and that the loss of tensile properties was not be related to the degradation of the material.

Degradation mechanism of the GFRP bar embedded in concrete environment

Based on the results of the tensile tests, SEM, FTIR and DSC, together with the results of earlier studies [10,11,18,21], a degradation mechanism of GFRP bar embedded in concrete beams is presented. The degradation process can be roughly divided into three stages, as follows.

(1) During the initial set of the concrete, the interstitial solution has a high alkalinity because of the cement hydration reaction. The OH^- ions and H_2O molecules penetrate into the internal of GFRP bar primarily through diffusion, the surface layer resin matrix of the GFRP bar swells up after absorbing water molecules, and then the osmotic pressure promotes the emergence and development of micro-cracks. (2) As the concrete gradually hardens and the carbonation reaction occurs, the moisture in concrete is consumed and evaporated. The corrosive reaction would stop in the absence of water. At this time, the concrete environment not only has no adverse effect on the tensile properties of the GFRP bars but also provides a protective cover against the erosion of outside aggressive media to the GFRP bars. (3) When the GFRP reinforced concrete beam is immersed in an alkaline solution to simulate the service condition, the GFRP bar is held in a state of tension under sustained flexural load. The initial defects and voids of resin matrix create a stress concentration and generate micro-cracks, these micro-cracks and the cracks in the concrete provide additional channels for the external aggressive media to attack the glass fibre and resin matrix.

As the moisture absorption increased, the weak interface layer induced cracks in the matrix. In addition, the different levels of expansion of the resin caused debonding and delamination of the interface, which resulted in a decrease in the stress transfer between the fibres and the matrix, and eventually lead to the degradation of the tensile properties of the GFRP bars. A schematic diagram of the degradation process is shown in Figure 9.

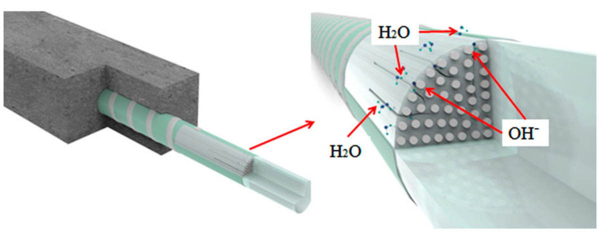

Figure 9. Schematic diagram for the degradation mechanism of GFRP bar embedded in concrete environment.

Conclusions

In this study, accelerated durability tests of GFRP bars embedded in concrete beams under service load with cracks were performed. After reaching the specified aging time, the bars were extracted from the beams. The tensile test was conducted to evaluate the residual tensile properties of the aged GFRP bars. In addition, SEM, FTIR and DSC were employed to characterise the deterioration of the glass fibre, matrix and fibre/matrix interface to further understand the degradation mechanisms of GFRP bars embedded in concrete beams. Based on the results obtained, the following conclusions may be drawn.

(i) The tensile strength of GFRP bars embedded in concrete beams in different environments decreased. After 18 months of alkaline solution exposure, the retention of the tensile strength is 79.7, 77.9 and 75.4% for the AS20, APS20 and APS40 specimens, respectively. Most of the tensile strength losses occurred during the initial stage. In contrast, almost no significant decrease was observed in the elastic modulus of the tested GFRP bars regardless of the environmental exposure or exposure period.

(ii) The microstructural features of the GFRP bar was observed after 18 months of aging, and no significant evidence of degradation was observed for sustained flexural loading less than 20% of the ultimate bending moment. When the sustained flexural load level increased to 40%, the glass fibre of the GFRP bars embedded in concrete beams with cracks exposed to an alkaline solution appeared to split.

(iii) The deterioration of the resin matrix was investigated using an FTIR and DSC analyses. FTIR analysis showed no irreversible changes in the polymer's chemical structure. Moreover, the DSC scans showed that a slight change in the T_g for the aged samples, which reveals little or no long-term effect of water or hydroxyl ingress on the bulk resin matrix material.

(iv) A crack in the concrete may increase the effect of the environment on the durability properties of the GFRP bar, but it cannot change the mechanisms of the mechanical degradation of the GFRP bar. In addition, the degradation rate of the GFRP bar was accelerated by sustained flexural loading.

(v) Compared with the deterioration of the glass fibre and resin matrix, debonding and delamination at the interface is more pronounced for GFRP bars embedded in concrete beams. The main reason for this is that less corrosive ions reach the inside of GFRP bar because the concrete provides a superior protective barrier.

Acknowledgements

The authors gratefully thank the Material Research and Testing Center of Wuhan University of Technology for their assistance and Professor Charles E. Bakis for his professional guidance.

Funding

This work was supported by National Nature Science Foundation of China [grant number 51178361].

References

[1] Alsayed SH, Alhozaimy AM, Al-salloum YA, et al. Durability of the new generation of GFRP rebars under severe environments. Proceedings Second International Conference on Durability of fiber reinforced polymer, 2002 May 29–31, p. 651–663.

[2] Zheng Y, Li C, Yu G, et al. Investigation of structural behaviors of laterally restrained GFRP reinforced concrete slabs. Compos B: Eng. 2012;43(3):1586–1597.

[3] Won JP, Lee SJ, Kim YJ, et al. The effect of exposure to alkaline solution and water on the strength-porosity relationship of GFRP rebar. Compos B: Eng. 2008;39(5):764–772.

[4] Chen Y, Davalos J, Ray l, et al. Accelerated aging tests for evaluations of durability performance of FRP reinforcing bars for concrete structures. Compos Struct. 2007;78:101–111.

[5] Robert M, Cousin P, Wang P, et al. Temperature as an accelerating factor for long-term durability testing of FRPS: should there be any limitations?. J Compos Constr. 2010;14(4):361–367.

[6] Micelli F, Nanni A. Durability of FRP rods for concrete structures. Constr Build Mater. 2004;18(7):491–503.

[7] Davalos JF, Chen Y, Ray I. Long-term durability prediction models for GFRP bars in concrete environment. J Compos Mater. 2011;46(16):1899–1914.

[8] Kamal A, Boulfiza M. Durability of GFRP rebars in simulated concrete solutions under accelerated aging conditions. J Compos Constr. 2011;15(4):473–481.

[9] Dejke V. Durability and service life prediction of GFRP for concrete reinforcement, 2001, Fibre-reinforced plastics for reinforced concrete structures, FRPRCS-5, Thomas Telford, 505–516.

[10] Almusullam TH, Al-Salloum YA. Durability of GFRP rebars in concrete beams under sustained loads at severe environments. J Compos Mater. 2006;40(7):623–637.

[11] Nkurunziza G, Benmokrane B, Debaiky AS, et al. Effect of sustained load and environment on long-term tensile properties of glass fiber-reinforced polymer reinforcing bars. ACI Struct J. 2005;102(4):615–621.

[12] He XJ, Yang JN, Bakis CE. Tensile strength characteristics of GFRP bars in concrete beams with work cracks under sustained loading and severe environments. J Wuhan Univ Technol. 2013;28(5):934–937.

[13] Robert M, Benmokrane B. Combined effects of saline solution and moist concrete on long-term durability of GFRP reinforcing bars. Constr Build Mater. 2013;38:274–284.

[14] Jia JH, Boothby TE, Bakis CE, et al. Durability evaluation of glass fiber reinforced-polymer-concrete bonded interfaces. J Compos Constr. 2005;9(4):348–359.

[15] Bakis CE. Durability of GFRP reinforcement bars, Advances in FRP composites in civil engineering – Proceedings 5th International Conference on FRP composites in civil engineering, 2011, CICE 2010, Beijing, 33–36.

[16] ACI Committee 440. Guide for design and construction of concrete reinforced FRP bars, 2003, Farmington Hills, MI, ACI 440.1R-03, American Concrete Institute.

[17] Mufti A. 'Durability of GFRP reinforced concrete in field structures', Proceedings 7th International Sym. on Fiber reinforced polymer reinforcement for reinforced concrete structures, 2005, American Concrete Institute, Detroit.

[18] Robert M, Cousin P, Benmokrane B. Durability of GFRP reinforcing bars embedded in moist concrete. J Compos Constr. 2009;13(2):66–73.

[19] Benmokrane B, Wang P, Ton-That TM. Durability of glass fiber-reinforced polymer reinforcing bars in concrete environment. J Compos Constr. 2002;6(3):143–153.

[20] Barkatt A. Issues in predicting long-term environmental degradation of fiber-reinforced plastics. In: Russel H. Jones, editor. Environmental effects on engineered materials. New York: CRC Press; 2001. p. 419–458.

[21] Zhu HG, Leung CKY, Kim J-K, et al. Degradation of glass fiber reinforced plastic composites containing nanoclay in alkaline environment. J Compos Mater. 2011;45(21):2147–2156.

[22] Montaigu M, Robert M, Ahmed EA, et al. Laboratory characterization and evaluation of durability performance of new polyester and vinylester E-glass GFRP Dowels for jointed concrete pavement. J Compos Constr. 2013;17(2):176–187.